

PROPOSAL FOR THE ESTABLISHMENT OF THE DATA SCIENCE AND ANALYTICS DIVISION

1.0 Strategic Rationale & Justification

1.1 Legal and Statutory Mandate

The creation of the Data Science and Analytics Division draws its statutory anchor directly from Section 5 of Legal Notice No. 273 of 2017 (under the Science, Technology and Innovation Act, No. 28 of 2013). This mandate legally obligates the institute to perform cutting-edge health and biological research alongside providing associated technical capacity support. In the modern research landscape, this mandate cannot be fully actualized without an independent, dedicated division to govern scientific data assets, model biological phenomena, and spearhead bioinformatics innovations.

1.2 Strategic Justification for Divisional Elevation

Elevating Data Science and Analytics from a standalone section to a full Division is critical due to several emerging institutional pressures:

- **Workload Scalability:** The volume, velocity, and complexity of data generated from advanced genomics, proteomics, and multi-entity biological surveillance projects have completely outgrown the capacity of a sub-section.
- **Specialization Demarcation:** Bioinformatics, disease modeling, and infrastructure automation require highly distinct scientific career pathways separate from traditional laboratory or administrative frameworks.
- **Data Safeguarding and Governance:** Institutional accountability mandates a central authority to govern biological and clinical data access, pipeline architecture, and privacy compliance.
- **Resource Mobilization:** A fully independent division strengthens the institute's position to secure international grant funding, establish corporate collaborations, and lead regional scientific consortiums.

2.0 Divisional Structure and Subsection Functions

To optimize operational efficiency and scientific focus, the Division is organized into three specialized subsections:

2.1 Biostatistics and Clinical Data Modeling Subsection

Responsible for standard research design, statistical oversight, and predictive modeling.

- Collect, analyze, and interpret biological and health science data across the institute.
- Build predictive models and machine-learning algorithms to discover disease trends.
- Evaluate the socio-economic effects of diseases and analyze costs/benefits of alternative control programs.

- Conduct institution-wide capacity building, training, and mentorship in biostatistics and modeling.

2.2 Bioinformatics and Genomics Analytics Subsection

Responsible for handling specialized molecular data, omics pipelines, and computational biological tools.

- Manage and process all genomic, proteomic, and transcriptomic data assets.
- Analyze complex genome sequencing projects and provide computational feedback to laboratory scientists.
- Design and deploy custom computational tools, algorithms, and automated analytical frameworks.
- Maintain rigorous up-to-date knowledge on global scientific literature regarding bioinformatics.

2.3 Data Infrastructure and Research Monitoring Subsection

Responsible for system pipelines, databases, and monitoring the analytical execution of projects.

- Establish data safeguarding mechanisms, security architectures, software integrations, and analytical workflows.
- Identify valuable external biological data sources and design automated pipelines for collection.
- Provide comprehensive monitoring and evaluation (M&E;) services to all research projects in the institute.
- Disseminate scientific findings and analytical insights through data portals, publications, and seminars.

3.0 Proposed Divisional Staffing Establishment

The following cadre framework outlines the necessary human resource expansion required to operationalize the new division safely and eliminate institutional progression bottlenecks:

Designation	Grade KIP	Approved	In Post	Proposed	Justification Summary
Director, Data Science & Analytics	2	0	0	1	To provide overall strategic leadership and policy direction.
Deputy Director / Chief Scientist	3	0	0	1	To manage inter-departmental operations and large research grants.
Assistant Director (by Subsection)	4	1	0	3	To provide technical management over the 3 new subsections.
Principal Research Scientists	5	0	0	3	To lead advanced pipeline designs and algorithmic research.
Senior Research Scientists	6	0	0	3	To execute complex modeling and genomic analytics workflows.

Research Scientists I/II	7/8	2	2	4	To execute day-to-day data cleaning and base programming tasks.
Total Divisional Cadre	-	3	2	15	To successfully operationalize the expanded statutory mandate.

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